

StockAI Agent

An Automated Stock-Trading Research System

A complete, plain-English guide for the total beginner

In one sentence

This is a computer program that automatically decides which stocks or funds to buy and sell, using fake (practice) money, so its ideas can be tested safely and scientifically before any real money is ever risked.

Two things it does

1) A momentum investor (the main strategy) - a hands-off system that buys a basket of funds once a month and holds them, meant for someone who wants to invest while working a normal job. **2) A day trader** (a research experiment) - a fast system that buys and sells the same 5 big-company stocks many times, used to study whether such trading can actually make money (spoiler: the honest answer is 'not reliably').

Built with the Rust programming language, packaged in Docker, using live market data from a broker called Alpaca. Every term in that sentence is explained inside.

1. What Is This Project, Really?

Imagine you want to grow your savings by buying and selling shares of companies, but you do not want to sit at a screen all day, and you definitely do not want to lose money while you are still learning. This project is a robot that does the buying and selling for you, following strict written rules, using **pretend money** so you can see whether its rules would have worked - without risking a single real dollar.

Think of it like a flight simulator for investing. A trainee pilot practises in a simulator before flying a real plane. This system 'practises' trading in a simulator (called **paper trading**) so we can measure, honestly and with real market prices, whether the strategy is any good.

The single most important idea in this whole project

Most people who build trading robots fool themselves - they see a few lucky wins and believe they are geniuses. This project was built specifically to AVOID fooling itself. It constantly compares its clever strategy against a 'dumb' strategy that just buys randomly (a coin flip). If the clever strategy cannot beat the coin flip, we admit it has no real skill. That honesty is the most valuable part of the whole system.

2. The Basic Vocabulary (read this first)

Before anything else, here are the everyday words of investing. Every later section relies on these.

Term	What it means (plain English)
Stock / Share	A tiny piece of ownership in a company. If you own 1 share of Apple, you own a microscopic slice of Apple. Its price goes up and down every second the market is open.
Ticker / Symbol	The short code for a stock. Apple = AAPL, Nvidia = NVDA, Microsoft = MSFT. Like a nickname the market uses.
ETF	Exchange-Traded Fund. A single 'basket' you can buy that holds many stocks at once. Example: buying 'SPY' means you own a slice of the 500 biggest US companies in one purchase. It trades like a normal stock.
Mutual fund	Also a basket of many stocks, but old-fashioned: it only prices ONCE per day (after the market closes), often charges penalties for selling quickly, and cannot be traded automatically. For that reason this project uses ETFs instead - they are the modern, tradeable equivalent.
Index	A scoreboard for a group of stocks. The 'S&P 500' index tracks the 500 largest US firms; the 'Nasdaq 100' tracks 100 big tech-heavy firms. You cannot buy an index directly, but you can buy an ETF that copies it (SPY copies the S&P 500; QQQ copies the Nasdaq 100).
Position	A holding you currently own. 'I have a position in NVDA' means 'I currently own some NVDA shares.'

Long	Betting a price will go UP by buying it. This whole system is 'long only' - it only ever buys hoping prices rise; it never bets on prices falling.
Portfolio	Everything you own added together - your stocks plus your uninvested cash.
Cash	Money not currently invested in anything, sitting idle and safe.
Buy / Sell / Entry / Exit	'Entry' = the moment you buy (enter a position). 'Exit' = the moment you sell (exit a position).
P&L	Profit and Loss - how much money you made or lost. '+\$5 P&L' means you gained five dollars.
Gross vs Net	Gross = profit BEFORE costs. Net = profit AFTER costs (fees, taxes). Net is what actually lands in your pocket, and it is always the honest number.

3. Trading Styles and Why Costs Matter So Much

Term	What it means (plain English)
Day trading	Buying and selling within the same day, often many times. Fast, exciting, and dominated by giant firms with supercomputers. Very hard for a small player to win at.
Swing trading	Holding a position for a few days to a few weeks. Slower, fewer trades, lower costs than day trading.
Investing	Holding for months or years. The slowest, cheapest, and historically most reliable way for ordinary people to grow money.
Fees / Costs	Every time you buy or sell, you lose a little money to costs: the 'spread', broker fees, and (on profits) taxes. These are small per trade but deadly if you trade often.
Spread	The tiny gap between the price to buy and the price to sell. You always buy a hair high and sell a hair low, so each round-trip quietly costs you a bit.
PDT rule	Pattern Day Trader rule (a US regulation). If your account is under \$25,000, you are only allowed 3 day-trades per 5 days. This is a big reason a \$500 day-trading account is impractical.

The costliest lesson this project proved with real data

With only \$500 and lots of fast trades, **costs ate about 137% of the gross profit** - meaning even when the trades won on paper, the fees turned it into a loss. This single fact is why the project moved away from fast day trading toward slow, low-cost investing. It is the maths of small accounts, and no amount of clever code changes it.

4. How We Measure 'Is It Any Good?'

Term	What it means (plain English)
Win rate	The percentage of trades that made money. Sounds important, but it is misleading on its own: you can win 20% of the time and still be rich if the wins are huge, or win 80% and go broke if the losses are huge.
Expectancy	The average profit per trade after weighing wins and losses. This is the honest measure of skill. Positive expectancy = the strategy makes money on average; negative = it loses on average.
Edge	A genuine, repeatable advantage that makes money beyond luck. The central question of the whole project: 'do we have an edge, or are we just getting lucky?'
Benchmark	The yardstick you compare against. If you cannot beat simply buying-and-holding a broad market ETF (like SPY), your strategy is not worth the effort. SPY and QQQ are used as benchmarks here.
Random baseline / Null hypothesis	A 'dumb' strategy that enters trades by coin flip, run side-by-side as a control. If the smart strategy cannot beat random coin flips, it has no real skill. This is the scientific heart of the project.
Overfitting	The classic self-deception: tuning a strategy so perfectly to past data that it looks amazing on history but fails on new data. The project fights this by pre-committing to rules and refusing to tweak based on tiny samples.

5. Market-Behaviour Terms Used by the Strategies

Term	What it means (plain English)
Volatility	How wildly a price jumps around. A calm stock barely moves; a volatile one swings a lot. High volatility = higher risk and bigger moves in both directions.
ATR	Average True Range. A number that measures a stock's typical daily price swing - basically a volatility ruler. If NVDA's ATR is \$3, it usually moves about \$3 a day. The system uses ATR to set wider safety-nets for wild stocks and tighter ones for calm stocks.
Momentum	The tendency of things that have been rising to keep rising for a while (and fallers to keep falling). One of the most studied, real effects in markets. The main strategy buys what has been going up.
Trend / Uptrend / Downtrend	The general direction of a price over time. Up = rising, down = falling. 'Do not fight the trend' - only buy when things are generally rising - is a core rule here.
Moving average / SMA	Simple Moving Average: the average price over the last N days, which smooths out the daily noise into a clean line. If today's price is above its 50-day average, the trend is considered up. Used as the market 'traffic light'.
Rolling average	An average that updates continuously over a sliding window of recent data. 'Rolling-average momentum' blends several time windows (1, 3, and 6 months) so one odd day cannot distort the signal.

Rebalancing	Periodically resetting your holdings back to the intended mix - e.g., once a month, sell what no longer qualifies and buy what now does. The momentum strategy rebalances monthly.
Regime	The overall 'weather' of the market - risk-on (generally rising, safe to buy) versus risk-off (falling/choppy, dangerous). The day trader now pauses buying in a risk-off regime.

6. The Technology (What It Is Built From)

You do not need to code to understand this, but here is what each piece is and why it exists.

Term	What it means (plain English)
Rust	A modern programming language known for being extremely fast and reliable. The 'brain' of the system (the part that makes trading decisions) is written in Rust so it can react quickly and never crash mid-decision.
Docker / Container	A way to package software so it runs identically on any computer, like a shipping container that fits any ship. The whole system runs as a set of Docker containers that start together.
Backend	The behind-the-scenes engine - it fetches prices, runs the strategies, and keeps score. It has no buttons; it just runs. (Port 8000, in tech terms.)
Frontend / Dashboard	The web page you look at in a browser to see what the system is doing (its holdings, its value). Runs on port 3000.
API	Application Programming Interface - a set of web addresses the backend offers so other programs (or you) can ask it questions, e.g. 'what are your current holdings?'
WebSocket	A live, always-open phone line between the price provider and the system, so prices stream in continuously instead of being asked for one at a time.
Alpaca	The broker and data provider. It supplies live and historical market prices and (in a real setup) would place the actual buy/sell orders. Here it is used in practice mode.
Redis	A super-fast temporary memory store used to pass data between parts of the system quickly.
ML sidecar	A helper program (in the Python language) running Machine Learning - software that learns patterns from data. It runs a model called 'Kronos'.
Kronos / Transformer	Kronos is an AI model of the 'transformer' type (the same family of AI behind modern chatbots), here trained to look at recent price charts and guess the next few price bars. It runs on a GPU (a graphics chip good at AI maths).

7. The Main Strategy: Monthly ETF Momentum Rotation

This is the strategy meant for real, hands-off use - the one a person with a day job would actually run. It is deliberately simple, slow, and low-cost.

How it works, step by step:

- **Once a month**, it looks at a menu of about 27 ETFs (baskets covering US stocks, foreign stocks, individual sectors like technology or energy, bonds, and gold).
- For each ETF it calculates **rolling-average momentum** - the blended average of how much that ETF rose over the past 1 month, 3 months, and 6 months. Blending three windows keeps one weird week from throwing off the ranking.

- It ranks them all and **buys the top 5**, splitting the money equally between them.
- **Safety switch (absolute momentum):** if one of the top 5 is actually falling (negative momentum), that slot is not bought - the money goes into a T-bill ETF called BIL (essentially cash) instead. This is what protects you in a market crash: when everything is falling, the strategy automatically hides in cash.
- It then **holds for the whole month** and does nothing until the next monthly check. Truly hands-off.
- Its value is compared every day against simply buying and holding SPY (the S&P 500), the honest benchmark.

Why this is the 'honest' strategy

Momentum investing on ETFs is one of the few approaches with genuine, decades-long academic evidence behind it. It will not make anyone rich overnight. Its realistic value is **lower stress, low fees, and smaller crashes** (thanks to the cash-safety switch), while roughly keeping pace with the market. That is exactly what a busy person with a second job actually needs.

8. The Research Experiment: The Day Trader

This is the fast system that trades 5 big US tech stocks (NVDA, AAPL, MSFT, GOOGL, AMZN). It is kept running as a living experiment. It scores each stock every second using 7 'signals', filters out weak ideas, sizes each bet, and manages exits carefully. Here is each part in plain English.

8a. The 7 Signals (each stock's 'opinions')

Every second, seven independent mini-analysts each give a score from strongly-sell to strongly-buy. Their scores are blended by importance (weight).

Term	What it means (plain English)
Kronos (weight 20%)	The AI transformer model predicting the next price bars. It can also 'veto' (block) a trade if it is bearish.
VP - Volume Profile (45%)	Looks at the price levels where the most shares actually changed hands today, to judge if a price is 'cheap' or 'expensive' relative to where the crowd traded. The most trusted signal - it can also veto.
Kalman filter (10%)	A mathematical smoother (borrowed from rocket guidance) that estimates the true underlying trend by filtering out random noise. Fires only on strong, sustained moves.
Pattern (5%)	Detects classic chart shapes and repeating patterns - the technical-analysis style of reading charts.
CVD - Cumulative Volume Delta (5%)	Tracks whether buyers or sellers are more aggressive by tallying buyer-initiated versus seller-initiated trades.
GEX - Gamma Exposure (weight 0%)	Estimates how options-market dealers are positioned, which can push prices. Currently switched off (weight 0) - it was unreliable.
COT - Commitment of Traders (0%)	A weekly report of what big institutions are betting. Too slow for fast trading, so switched off.

8b. The Filter Gauntlet (an idea must survive every gate)

Before any stock is bought, it must pass through a series of veto gates. Failing any one gate cancels the trade.

- **Market regime filter:** if the overall market (QQQ) is below its 50-day average, the whole day trader pauses - no new buys in a falling market. (This was a recent major improvement.)
- **Cooldown:** will not re-buy the same stock within 3 hours, to avoid frantic churning.
- **Kronos veto:** the AI must not be predicting a fall.
- **Trend veto:** the stock's price must be above its own intraday average (do not buy something falling right now).
- **Consensus veto:** if 4 or more of the 7 signals are bearish, skip it.

- **VP veto:** if Volume Profile says clearly overpriced, skip it.
- **Score threshold:** the blended score must clear a minimum bar of conviction.

8c. Position Sizing (how much to buy)

The system splits its available cash across open 'slots' (max 5 positions), never putting more than 25% into any one stock, and it bets bigger when its confidence score is higher (full size for strong signals, half size for weak ones). No human picks the dollar amount - it is pure arithmetic.

8d. The Exit Ladder (when to sell - the part that actually works)

Selling is decided by a priority list checked every second. The newest upgrade made these **ATR-based**, meaning each stock's safety-nets are sized to its own volatility - wide for wild stocks, tight for calm ones.

- **Hard stop** (about 2x ATR loss): emergency exit to protect capital immediately.
- **Take profit** (about 3x ATR gain): lock in a solid win.
- **Partial booking:** sell half at a smaller gain (about 1.5x ATR) and let the rest 'run' for more.
- **Trailing stop** (about 1.5x ATR from the peak): once in profit, sell if the price falls back a set amount from its high, protecting gains.
- **Bearish exit / time-out:** exit if the signals clearly turn negative, or after holding too long with nothing happening.

9. How We Know If Any Of This Works

This is what makes the project genuinely scientific rather than wishful thinking.

Term	What it means (plain English)
Shadow traders	Extra copies of the strategy running invisibly in parallel on the same live prices, each with one setting changed. They let us test 'what if we did X instead?' without risking the main account.
The random baseline	One shadow trader ignores all signals and buys by coin flip. It is the bar every clever strategy must beat. If clever cannot beat random, clever has no skill.
The edge test	A report (run by a script called analyze.ps1) that ranks the real strategy against all the shadow traders and the random baseline, measuring expectancy per trade.
Kill criterion	A pre-agreed deadline and rule: 'beat the benchmark by this date or admit the idea is dead.' Deciding this in advance is what stops us from moving the goalposts to protect our ego.

The verdict the data actually delivered

After a month of live paper trading, the honest result was clear and humbling: **the 7-signal day trader lost to the random coin-flip baseline**. The clever filters were not adding skill - if anything they were removing good luck. This is not a failure of the project; it is the project succeeding at its real job: telling the truth. Beating the market with a small automated account is extraordinarily hard, and the system was honest enough to prove it.

10. The Journey (Why the Project Changed Over Time)

- **Started as a fast day trader** on 5 tech stocks - and discovered fees devour tiny accounts (137% of gross).
- **Slowed to swing trading** (holding days, fewer trades) to cut those fees and dodge the PDT rule.
- **Tested honestly and found no edge** - the signals lost to random coin flips.
- **Pivoted to the momentum ETF strategy** - the low-touch, evidence-backed approach that fits real life.
- **Kept the day trader alive as a research lab** and kept improving its genuinely useful parts: a market-regime filter (do not buy in falling markets) and ATR-based exits (risk sized to each stock's volatility).

The throughline: instead of chasing a fantasy of easy riches, the project kept following the evidence, and the evidence kept pointing toward slow, diversified, low-cost, rules-based investing - with rigorous honesty about what does and does not work.

11. Living With the System Day to Day

- **The dashboard** (a web page) opens on the Momentum Portfolio - showing its holdings, its value, and how it compares to SPY. The old day trader is tucked into a secondary 'legacy' tab.
- **Persistence:** the system saves its state to disk constantly, so if the computer is turned off overnight, it resumes exactly where it left off instead of forgetting everything.
- **The one human job:** keep the computer awake during market hours (9:30am-4:00pm US Eastern) so it can update. Because the main strategy only acts monthly, missing a day here or there costs nothing.
- **Reading results:** a report script summarises performance and the all-important edge test any time you run it.

The bottom line, honestly stated

This is a genuinely well-engineered system that does something rare: it refuses to lie to its owner. It will not make you rich quickly - nothing safely does. Its real value is twofold: (1) a sensible, low-effort momentum-investing strategy you can actually run alongside a job, and (2) a rigorous, honest laboratory that proves, with data, which trading ideas hold up and which are just wishful thinking.

12. Quick-Reference: Every Abbreviation in One Place

Term	What it means (plain English)
AI	Artificial Intelligence - software that learns patterns from data.
API	Application Programming Interface - web addresses to ask the system questions.
ATR	Average True Range - a measure of a stock's typical daily price swing (volatility ruler).
BIL	An ETF holding ultra-safe short-term US government T-bills; used as the 'cash / safety' hiding place.
COT	Commitment of Traders - a weekly report of big-institution positioning (signal, currently off).
CVD	Cumulative Volume Delta - buyer vs seller aggression measure (signal).
EOD	End Of Day - the market close, when daily wrap-up reports are produced.
ETF	Exchange-Traded Fund - a tradeable basket of many stocks.
GEX	Gamma Exposure - options-dealer positioning measure (signal, currently off).
GPU	Graphics Processing Unit - a chip that is very good at AI maths; runs the Kronos model.
ML	Machine Learning - software that learns from data (the 'sidecar' helper).
NAV	Net Asset Value - the once-a-day price of a mutual fund.
P&L	Profit and Loss - money made or lost.
PDT	Pattern Day Trader rule - US law limiting day trades on accounts under \$25,000.
QQQ	The ETF that tracks the Nasdaq-100 (big tech index).
SMA	Simple Moving Average - the smoothed average price over N days (trend line).
SPY	The ETF that tracks the S&P 500 (the 500 biggest US firms); the main benchmark.
T-bill	Treasury Bill - an ultra-safe short-term loan to the US government; the safest 'cash' asset.
VP	Volume Profile - where the most shares traded, used to judge cheap vs expensive (top signal).

Document generated automatically from the live project. Every strategy, filter, and term described here corresponds to real code in the system.